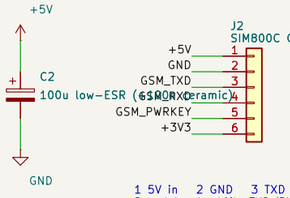
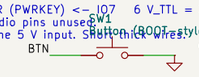


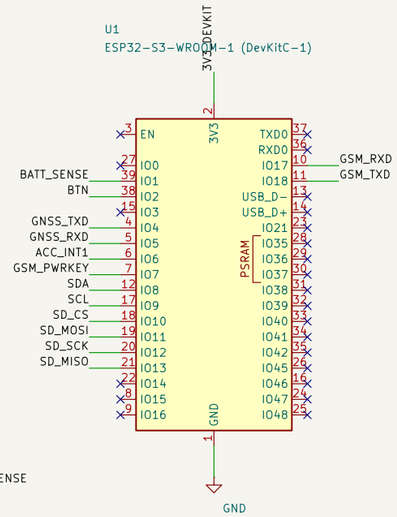
1 VCC 5V 2 GND 3 TX -> IO4 4 RX <- IO5
5 SCL / 6 SDA: compass (Foxeer only, 0x0D). BE-220 has pins 1-4.

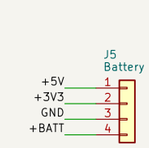


1 5V in 2 GND 3 TXD -> IO18 4 RXD <- IO17 5 PWR (PWRKEY) <- IO7 6 V_TTL = 3V3
Board level-shifts TXD/RXD to V_TTL; VBAT, DTR, RI, VRTC, audio pins unused.
Board carries 820 uF 6.3 V on the modem rail; C2 steadies the 5 V input. Short thick wires.

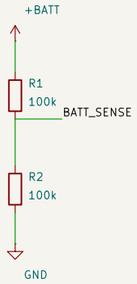


Internal pull-up on IO2; GPIO interrupt on both edges.



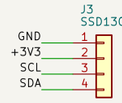


1 5V out (<= 2.2 A cont.) 2 3V out (1 A; measure: 3.0 or 3.3 V) 3 GND 4 B+ (cell)
Feeds: +5V -> devkit 5V pin, GNSS, SIM800C board. +3V3 -> OLED, SD, BMA400.

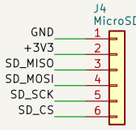


Cell 4.2 V max -> 2.1 V at IO1 (ADC1_CH0, 12 dB).

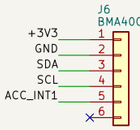
DevKitC-1: 5V pin <- +5V rail (onboard LDO makes 3V3_DEVKIT).
USB, EN, BOOT, JTAG and the onboard RGB LED (IO38/IO48) stay on the devkit.
SuperMini build: GSM on IO43/IO44 instead of IO17/IO18.



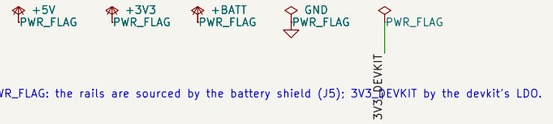
1 GND 2 VCC 3V3 3 SCL <- IO9 4 SDA <> IO8
Module pull-ups on SDA/SCL; bus at 100 kHz.



1 GND 2 VCC 3V3 (or 5V if the module has a level shifter)
3 MISO -> IO13 4 MOSI <- IO11 5 SCK <- IO12 6 CS <- IO10



1 VCC 3V3 2 GND 3 SDA 4 SCL 5 INT1 -> IO6 (RTC GPIO, deep-sleep wake) 6 INT2 n/c
I2C 0x14 (0x15 with SDO high).



PWR_FLAG: the rails are sourced by the battery shield (J5); 3V3_DEVKIT by the devkit's LDO.

Breakout modules drawn as pin headers; see docs/HARDWARE.md
ESP32-S3 DevKitC-1 development build; pins per include/config.hpp

Maksym Kutyshenko

Sheet: /
File: gps_tracker.kicad_sch

Title: GPS Tracker

Size: A3	Date: 2026-10-09	**Rev: 1**
KiCad E.D.A. 10.0.4		Id: 1/1